



NUMA

Multimodal Semantic Neural Decoding · EEG + fNIRS

Reading which of 36 imagined objects a person is thinking —
from non-invasive brain signals.

Neel Ahuja · CS 153 — The One-Person Frontier Lab

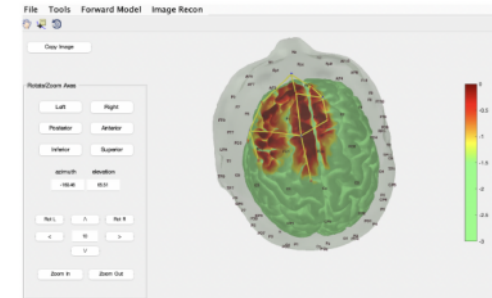
Live demo: numa-frontier-systems.vercel.app



Non-invasive brain decoding is bottlenecked by signal-to-noise.

- ▶ The skull smears the brain's weak electric fields → EEG is fast but spatially blind.
- ▶ Decode which of 36 imagined objects (animals/tools), single-trial, non-invasive. Chance = 2.78%.
- ▶ Insight: fuse EEG (millisecond timing) with fNIRS (blood flow → strong spatial localization).

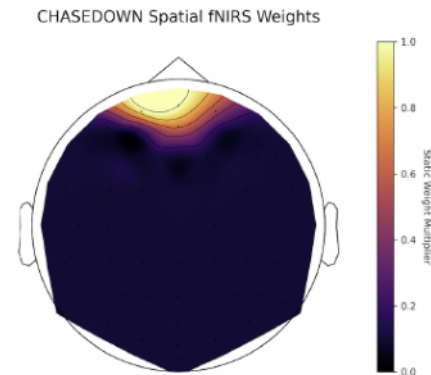
fNIRS cortical sensitivity



A physics-informed fNIRS prior on EEG

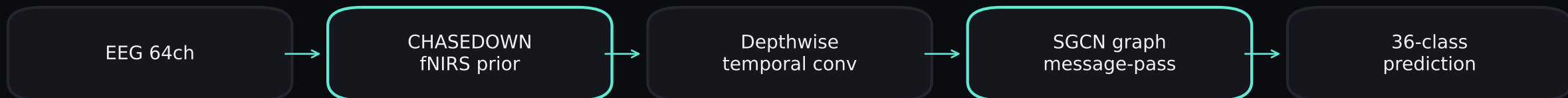
- ▶ EEG → high temporal resolution, low spatial resolution
- ▶ fNIRS → high spatial resolution, slow (hemodynamic lag)
- ▶ CHASEDOWN: fNIRS blood-flow map → per-electrode spatial weight — zero trainable parameters.
- ▶ Inject into EEG: $E_{\text{fused}}(t) = W_{\text{static}} \odot E_{\text{raw}}(t)$

CHASEDOWN static prior (front-left)



Topology-aware decoding (SGCN)

- ▶ Depthwise spatio-temporal backbone (ENIGMA): separates WHEN a rhythm occurs from WHERE — few params, noise-robust.
- ▶ SGCN models 64 electrodes as a graph from true 10-20 distances: localized message passing along the cortex.
- ▶ The fNIRS prior is routed along real cortical pathways instead of a flat grid — the key to making fNIRS help.



fNIRS + EEG significantly outpaces EEG alone

Avg. accuracy, 5-fold cross-validation

Chance 2.78%

EEG only (ENIGMA baseline) 4.8%

EEG + fNIRS prior, no graph (naive) 2.93%

EEG + fNIRS + SGCN (multimodal) 9.16%

+90.8%

relative gain over EEG-only

p = 0.005

statistically significant

Naive fusion (no graph): $p = 0.1357$
→ fNIRS only helps WITH
topological routing.

Autoresearch: an AI agent that improves the model

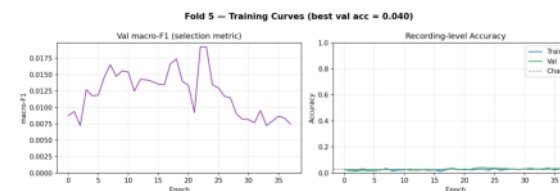
- ▶ An autonomous LLM agent edits train.py, runs the experiment, measures, and keeps or reverts the change — in a loop.
- ▶ Frozen, un-gameable benchmark: fixed split, fixed time budget, scored on collapse-proof balanced accuracy.
- ▶ 26 experiments → 4 confirmed winners, 22 dead-ends ruled out (capacity, extra layers, augmentations, attention collapse).
- ▶ Turns one researcher into an automated experimental program.



Six model generations, honestly evaluated

- ▶ v3 — class-balanced SGCN (kills single-class collapse)
- ▶ v4 — autoresearch-optimized config (the 4 winners)
- ▶ v5 — EEGNet backbone + spectral filter-bank fusion
- ▶ v6 — recording-level ATTENTION model: learns which 1-s actually carry the concept (deployed model).

v6 training dynamics

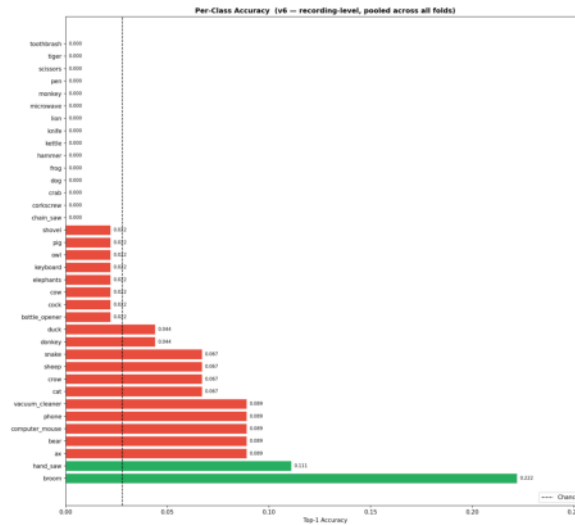
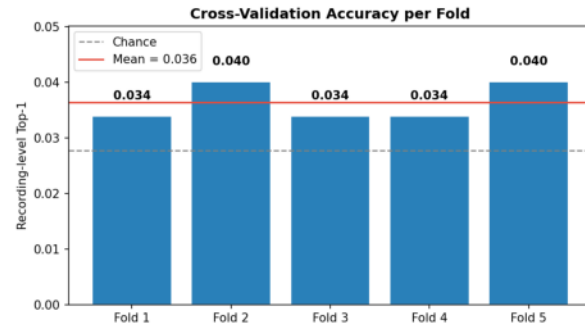


Cross-Validation Accuracy per Fold

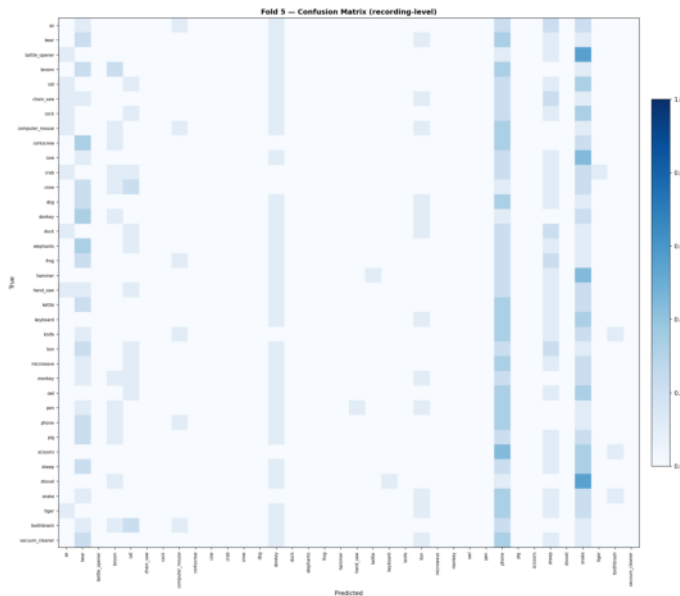
Recording-level Top-1

--- Chance
— Mean = 0.036

Fold	Recording-level Top-1 Accuracy
Fold 1	0.034
Fold 2	0.040
Fold 3	0.034
Fold 4	0.034
Fold 5	0.040



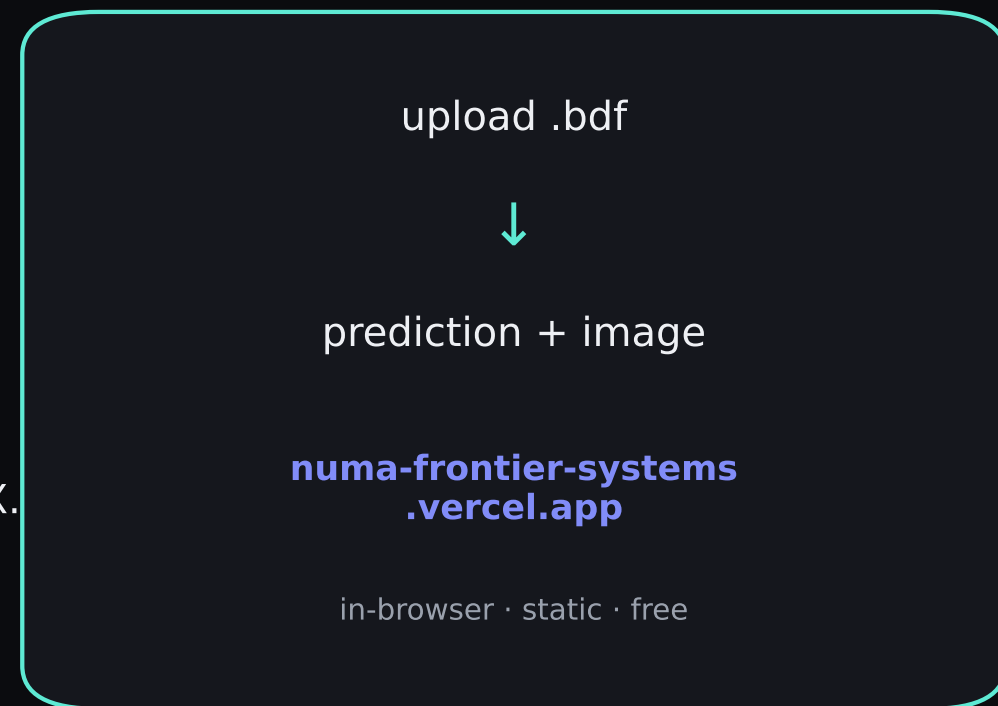
Honest reporting & failure analysis



- ▶ Leave-one-instance-out CV → no data leakage.
- ▶ Balanced accuracy throughout — can't be faked by guessing one common class.
- ▶ Limitation surfaced: scarce data → the 36-class manifold can collapse → motivates more data.
- ▶ Single-trial 36-way EEG is hard; the win is consistent collapse-free separation above chance.

A live, in-browser decoder anyone can try

- ▶ Upload one .bdf recording → predicted object pops up with a top-5 confidence chart.
- ▶ Runs 100% in the browser (onnxruntime-web) — no server, no upload limits, data never leaves your device.
- ▶ Static deploy on Vercel; ~118K-parameter model (<1 MB).
- ▶ Full JS pipeline: parse → filter → window → band-power → ONNX.



Why it matters

- ▶ A foothold toward communication for people who have lost speech and movement — intention-driven interfaces.
- ▶ Evidence that physics-informed multimodal fusion + topology-aware models is the right direction for low-SNR BCI.
- ▶ Proof that one person + agentic tooling can run a full research program: search → train → evaluate → deploy.

Roadmap

- ▶ A larger synchronous fNIRS-EEG cohort to escape the data-starved regime that causes collapse.
- ▶ A trainable fusion that learns the fNIRS→EEG mapping instead of a static prior.
- ▶ Stimulus-onset-locked epoching — the biggest remaining lever on the EEG-only ceiling.
- ▶ Fully on-device multimodal inference in the browser.

AI use, credits & reproducibility

- ▶ AI disclosure: the autoresearch loop IS an autonomous LLM agent; AI assistants also helped implement, iterate, and document.
- ▶ All claims backed by 5-fold CV numbers and a significance test — not by model assertion.
- ▶ Builds on ENIGMA (Kneeland et al., 2026) and the open dataset Rybář et al., 2025 — both cited.
- ▶ Public repo + commit history + all model versions + live demo.

Dataset: openneuro.org/datasets/ds004514 · ENIGMA: [arXiv:2602.10361](https://arxiv.org/abs/2602.10361)

NUMA

fNIRS + EEG, routed through cortical topology,
beats EEG alone — verified, and shipped.

Try it: numa-frontier-systems.vercel.app

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